

Leach Pad Optimization with Smart Worker

01

Cynide flow rates

Operators capture flow rates using the Smart Worker app. It can be used in offline mode.

02

Data compilation

The information captured in the leach pad along with laboratory cells grades are compiled in a Excel Spreadshet.

03

Diversion of solution streams

The streams are automatically diverted using Solver to maximize performance.

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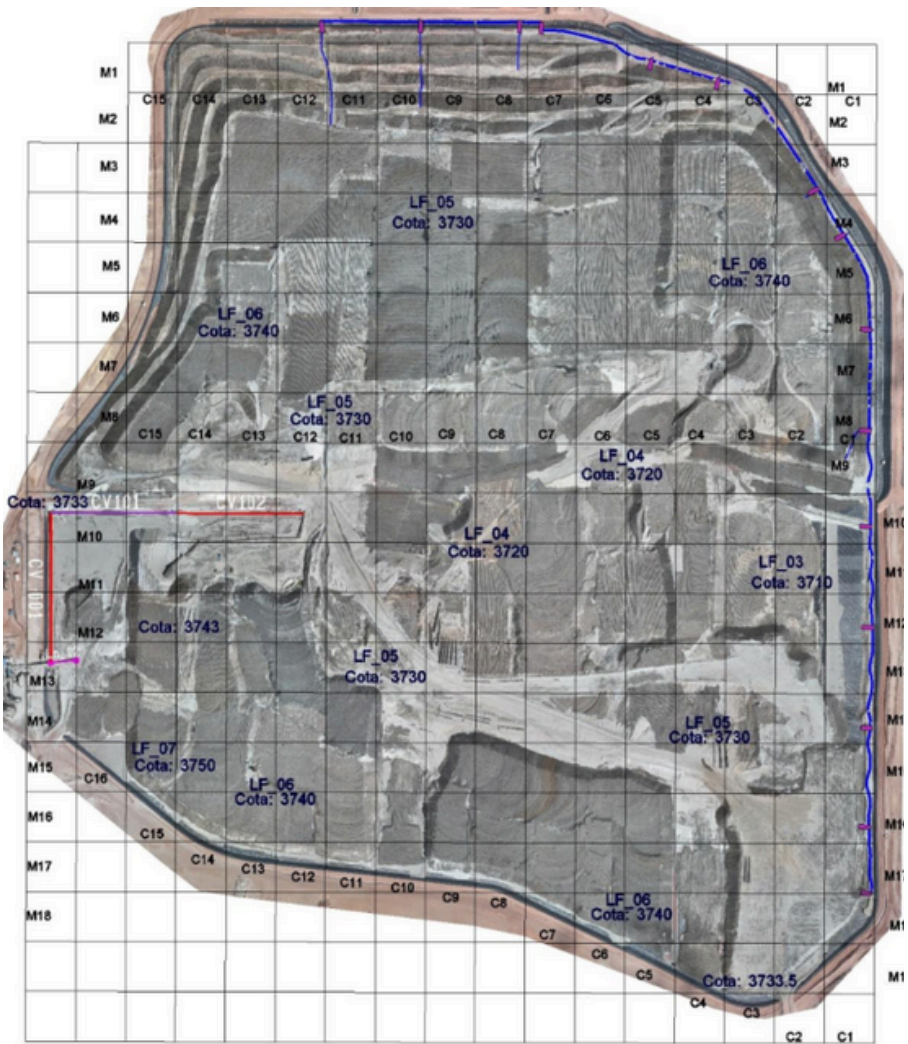
Cyanide flow rates

About Smart Worker

The Cyanide Flow Rate Registration App enables operators to digitally capture real-time solution flow rates from each leach pad cell.

Key Features

- Full offline functionality, essential due to the leach pad's distance and minimal connectivity
- Recording of flow rates, cell ID, lift, solution type, and relevant comments
- Dynamic input validation to detect inconsistent or unrealistic flow readings
- Automatic sync to Microsoft 365 once a connection is available
- Compatibility with optimization engines



Leach Pad Grid Map



Capturing flow rates using a mobile device

Smart Worker UI

01. Cell, area and lift selection

02. Flow rate register and submission

Smart Worker

Selecciona las áreas de riego

Introduce el área de riego

Área 1, Celda, Lift

Area

Celda

Lift

Área 2, Celda, Lift

Area

Celda

Lift

Área 3, Celda, Lift

Area

Celda

Área 4, Celda, Lift

Area

Celda

Lift

Área 5, Celda, Lift

Area

Celda

Lift

Observaciones

OK

Smart Worker

Áreas seleccionadas: A117 A122

Introduce la tasa de riego (mL/s)

Introduce caudal (m3/h)

22.79

Observaciones

Enviar

Data Compilation

Excel Version

The data is compiled in an Excel spreadsheet. Excel was deliberately chosen due to its user-friendly interface; therefore, this tool is accessible to any Leach Pad supervisor. Table 1 presents a summary of the relevant information provided by Smart Worker (2) and the laboratory (3). Its clear and straightforward format makes the optimization process practical and effective.

01

	LEACH PAD			LABORATORY	PROCESSING PLANT	
	Flujo m3/h					
Celda	PLATFORM	SLOPE	TOTAL	GRADE	POND	OUNCES ESTIMATION
C1	-	-	-			
C2	22.79	-	22.79	0.21	1.00	3.69
C3	52.03	-	52.03	1.11	1.00	44.57
C4	59.01	-	59.01	0.25	1.00	11.38
C5	-	-	-			
C6	-	-	-			
C7	-	6.47	6.47	0.25	1.00	1.25
C8	296.01	18.81	314.82	0.37	0.00	89.89
C9	167.52	10.01	177.53	0.20	1.00	27.40
C10	122.75	7.29	130.04	0.20	0.00	20.07
C11	98.50	5.90	104.39	0.19	0.00	15.31
C12	24.23	6.39	30.61	0.18	0.00	4.25
C13	129.19	-	129.19	0.20	0.00	19.94
C14	-	-	-			
C15	5.00	-	5.00	0.27	0.00	1.04
C16	5.00	-	5.00	0.41	1.00	1.58
TOTAL	982.04	54.86	1,036.89			240.38

Data compilation (Laboratory + Flow rates)

02

Editor avanzado

cr5c2_sw_tasas

```
let
    Origen = CommonDataService.Database("orgc53b46fa.crm2.dynamics.com"),
    dbo_cr5c2_sw_tasas = Origen{[Schema="dbo",Item="cr5c2_sw_tasas"]}[Data],
    #"Columnas quitadas" = Table.RemoveColumns(dbo_cr5c2_sw_tasas,{"cr5c2_sw_tasasid", "createdby", "modifiedon", "modifi
    #"Columnas con nombre cambiado" = Table.RenameColumns(#"Columnas quitadas",{{"cr5c2_id", "id"}, {"cr5c2_area", "area"
    #"Columnas quitadas1" = Table.RemoveColumns(#"Columnas con nombre cambiado",{"fecha"})
in
    #"Columnas quitadas1"
```

Leach Pad data retrieval (Dataverse)

03

Base de datos SQL Server

Servidor ⓘ

Base de datos

▲ Opciones avanzadas

Tiempo de espera del comando en minutos (opcional)

Instrucción SQL (opcional, requiere una base de datos)

SELECT [Muestra]
 ,[T_EXTERNAL_ID]
 ,[Fecha_muestra]
 ,[Análisis]
 ,[Valor]
 ,[Analito]

☒ Incluir columnas de relación

☐ Navegar usando la jerarquía completa

☐ Habilitar la compatibilidad con la conmutación por error de SQL Server

Laboratory data retrieval (SQL Server)

Diversion of solution streams

Leach Pad Optimization

The objective is to maximize the number of ounces routed to the processing plant. This is achieved using Solver. Solution streams can be diverted either to the PLS or the ILS pond. Each pond feeds different tanks in the ADR plant.

The efficiency of each system varies, encouraging supervisors to route higher flows to the more efficient system. However, both systems are subject to capacity constraints, and each cell contains a different gold grade.

Solver iterates to determine the optimal diversion strategy that maximizes gold ounces.

0 → ILS

1 → PLS

Parámetros de Solver

Establecer objetivo:

\$O\$14

Para:

☒ Máx

☐ Mín

☐ Valor de:

0

Cambiando las celdas de variables:

\$F\$4:\$F\$19

Sujeto a las restricciones:

\$F\$4:\$F\$19 = binario

\$M\$5 <= 470

\$N\$5 <= 400

Agregar

Cambiar

Eliminar

Restablecer todo

Cargar/Guardar

☐ Convertir variables sin restricciones en no negativas

Método de resolución:

Simplex LP

Opciones

Solver settings

ADR PLANT	1	0
Flow rate (m3/h)	323	399
Grade (ppm)	0.36	0.11
Barren (pm)	0.03	0.05
Oz Estim	119.97	18.52

Total Oz Estim		138.49
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Flow rate constraints (m3/h)	470	400
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Final ounces estimation